

Ciprinol *Ciprofloxacin*

1. NAME OF THE MEDICINAL PRODUCT

Ciprinol 500 mg film-coated tablets

2. QUALITATIVE AND QUANTITATIVE COMPOSITION

Each film-coated tablet contains 500 mg ciprofloxacin as ciprofloxacin hydrochloride monohydrate.

The 500 mg film-coated tablets are white, oval, scored on one side. The tablet can be divided into equal halves.

3. CLINICAL PARTICULARS

3.1 Therapeutic indications

Ciprinol film-coated tablets are fluoroquinolone antibacterial agent indicated for the treatment of the following infections. Special attention should be paid to available information on resistance to ciprofloxacin before commencing therapy.

Consideration should be given to official guidance on the appropriate use of antibacterial agents.

Uncomplicated and complicated infections caused by ciprofloxacin susceptible pathogens.

Infections of the respiratory tract:

Ciprofloxacin can be regarded as an advisable treatment for pneumonias caused by *Klebsiella*, *Enterobacter*, *Proteus*, *E. coli*, *Pseudomonas*, *Haemophilus*, *Moraxella catarrhalis*, *Legionella* and *Staphylococcus*.

Infections of the middle ear (otitis media), of the paranasal sinuses (sinusitis), especially if these are caused by Gram-negative organisms including *Pseudomonas aeruginosa* or by staphylococci.

Infections of the eyes

Infections of the kidneys and/or the efferent urinary tract

Infections of the genital organs, including adnexitis, gonorrhea, prostatitis

Infections of the abdominal cavity (e.g. infections of the gastrointestinal tract or of the biliary tract, peritonitis)

Infections of the skin and soft tissue

Infections of the bones and joints

Sepsis

Infections or imminent risk of infection (prophylaxis) in patients whose immune system has been weakened (e.g. patients on immunosuppressants or have neutropenia)

Selective intestinal decontamination in immunosuppressed patients

Prophylaxis of invasive infections due to *Neisseria meningitidis*.

Children and adolescents

Ciprofloxacin may be used in children for the second- and third-line treatment of complicated urinary tract infections and pyelonephritis due to *Escherichia coli* (age range applied in clinical studies: 1 – 17 years) and for the treatment of broncho-pulmonary infections of cystic fibrosis associated with *Pseudomonas aeruginosa* (age range applied in clinical studies: 5 – 17 years).

Treatment should only be initiated after careful benefit/risk evaluation, due to possible adverse events related to joints and/or surrounding tissues.

The clinical trials in children were performed in the indications listed above. For other indications clinical experience is limited.

Inhalational anthrax (post-exposure) in adults and in children

To reduce the incidence or progression of disease following exposure to aerosolized *Bacillus anthracis*.

3.2 Posology and method of administration

Posology

Unless otherwise prescribed, the following daily doses are recommended for

Adults

Table 1: Recommended daily doses of Ciprinol film-coated tablets in adults

Indications	Daily dose of ciprofloxacin in mg for Ciprinol film-coated tablets
Infections of the respiratory tract I (according to severity and organism)	2 x 500 mg to 2 x 750 mg
Urinary tract infections: - acute, uncomplicated - cystitis in women (before menopause) - complicated	2 x 250 mg to 2 x 500 mg single dose 500 mg 2 x 500 mg to 2 x 750 mg
Genital infections - uncomplicated gonorrhea (including extragenital sites of infection) - adnexitis, prostatitis, epididymo-orchitis	1 x 500 mg 2 x 500 mg to 2 x 750 mg
Diarrhea	2 x 500 mg
Other infections (see indications)	2 x 500 mg
Particularly severe, life-threatening infections, i.e. - Recurrent infections in cystic fibrosis - Bone and joint infections - Septicemia - Peritonitis.	2 x 750 mg

In particular when <i>Pseudomonas</i> , <i>Staphylococcus</i> or <i>Streptococcus</i> is present	
Inhalational anthrax (post exposure)	2 x 500 mg
Prophylaxis of invasive infections due to <i>Neisseria meningitidis</i>	1 x 500 mg as a single dose

Children and adolescents

Table 2: Recommended daily doses of Ciprofloxacin oral in children and adolescents

Indications	Daily dose of ciprofloxacin in mg for Ciprinol film-coated tablets
Infections in cystic fibrosis	2 x 20mg/kg body weight (maximum of 750 mg per dose)
Complicated urinary tract infections and pyelonephritis	2 x 10mg/kg body weight to 2 x 20mg/kg body weight (maximum of 750 mg per dose)
Inhalational anthrax (post-exposure)	2 x 15mg/kg body weight (maximum of 500mg per dose)

Missed dose

If a dose is missed, it should be taken as soon as the patient remembers and then treatment should be continued as prescribed. Double doses should not be taken to compensate for a missed dose.

Additional information on special patient population

Children and adolescents

For recommended dosage regimen see Table 2.

Geriatric patients (> 65 years)

Elderly patients should receive a dose as low as possible depending on the severity of their illness and the creatinine clearance. (see also 'Patients with renal and hepatic impairment')

Patients with renal and hepatic impairment

Adults

- Patients with renal impairment

Table 3: Recommended doses for patients with renal impairment

Creatinine Clearance [mL/min/1.73 m ²]	Serum Creatinine [mg/100mL]	Total daily oral dose of ciprofloxacin
30 to 60	1.4 to 1.9	maximum 1000 mg
below 30	≥2.0	maximum 500mg

- Patients with renal impairment on hemodialysis
- For patients with creatinine clearance between 30 and 60 mL/min/1.73m² (moderate renal impairment) or serum creatinine concentration between 1.4 and 1.9 mg/100 mL, the maximum daily oral dose of ciprofloxacin should be 1000 mg or 800 mg for an intravenous regimen.

-For patients with creatinine clearance less than 30 mL/min/1.73m² (severe renal impairment) or serum creatinine concentration equal or higher than 2.0 mg/100 mL, the maximum daily oral dose of ciprofloxacin should be 500 mg or 400 mg for an intravenous regimen on dialysis days after dialysis

- Patients with renal impairment on continuous ambulatory peritoneal dialysis (CAPD)
- Addition of Ciprofloxacin solution for infusion to the dialysate (intraperitoneal): 50 mg ciprofloxacin / liter dialysate administered 4 times a day every 6 hours
- The maximum daily oral dose of ciprofloxacin should be (1 x 500 mg Ciprofloxacin film-coated tablet or 2 x 250 mg Ciprofloxacin film-coated tablets).

- Patients with hepatic impairment
- In patients with impaired hepatic function no dose adjustment is required

- Patients with renal and hepatic impairment
- For patients with creatinine clearance between 30 and 60 mL/min/1.73m² (moderate renal impairment) or serum creatinine concentration between 1.4 and 1.9 mg/100 mL, the maximum daily oral dose of ciprofloxacin should be 1000 mg or 800 mg for an intravenous regimen.
- For patients with creatinine clearance less than 30 mL/min/1.73m² (severe renal impairment) or serum creatinine concentration equal or higher than 2.0 mg/100 mL, the maximum daily oral dose of ciprofloxacin should be 500 mg or 400 mg for an intravenous regimen.

Children and adolescents

Dosing in children with impaired renal and or hepatic function has not been studied.

Inhalational Anthrax (Post-exposure) in Adults and Children

For recommended dosage regimen see Table 1 (for adults) and Table 2 (for children and adolescents), respectively.

Method of administration

Tablets are to be swallowed unchewed with fluid. They can be taken independent of mealtimes. If taken on an empty stomach, the active substance is absorbed more rapidly. Ciprofloxacin tablets should not be taken with dairy products (e.g. milk, yoghurt) or mineral-fortified fruit-juice (e.g. calcium-fortified orange juice) (see section 3.5).

In severe cases or if the patient is unable to take tablets (e.g. patients on enteral nutrition), it is recommended to commence therapy with intravenous ciprofloxacin until a switch to oral administration is possible.

Route of administration

For oral use.

3.3 Contraindications

- Hypersensitivity to the active substance, to other quinolones or to any of the excipients (see section 2).
- Concomitant administration of ciprofloxacin and tizanidine (see section 3.5).

3.4 Special warnings and precautions for use

The use of ciprofloxacin should be avoided in patients who have experienced serious adverse reactions in the past when using fluoroquinolones containing products (see section Undesirable Effects). Treatment of these patients with ciprofloxacin should only be initiated in the absence of alternative treatment options and after careful benefit/risk assessment.

Severe infections and mixed infections with Gram-positive and anaerobic pathogens

Ciprofloxacin monotherapy is not suited for treatment of severe infections and infections that might be due to Gram-positive or anaerobic pathogens. In such infections ciprofloxacin must be co-administered with other appropriate antibacterial agents.

Streptococcal Infections (including *Streptococcus pneumoniae*)

Ciprofloxacin is not recommended for the treatment of streptococcal infections due to inadequate efficacy.

Genital tract infections

Epididymo-orchitis and pelvic inflammatory diseases may be caused by fluoroquinolone-resistant *Neisseria gonorrhoeae*. Ciprofloxacin should be co-administered with another appropriate antibacterial agent unless ciprofloxacin-resistant *Neisseria gonorrhoeae* can be excluded. If clinical improvement is not achieved after 3 days of treatment, the therapy should be reconsidered.

Intra-abdominal infections

There are limited data on the efficacy of ciprofloxacin in the treatment of post-surgical intra-abdominal infections.

Travellers' diarrhoea

The choice of ciprofloxacin should take into account information on resistance to ciprofloxacin in relevant pathogens in the countries visited.

Infections of the bones and joints

Ciprofloxacin should be used in combination with other antimicrobial agents depending on the results of the microbiological documentation.

Inhalational anthrax

Use in humans is based on *in-vitro* susceptibility data and on animal experimental data together with limited human data. Treating physicians should refer to national and/or international consensus documents regarding the treatment of anthrax.

Children and adolescents

The use of ciprofloxacin in children and adolescents should follow available official guidance. Ciprofloxacin treatment should be initiated only by physicians who are experienced in the treatment of cystic fibrosis and/or severe infections in children and adolescents.

Treatment should be initiated only after a careful benefit/risk evaluation, due to possible adverse events related to joints and/or surrounding tissue.

Broncho-pulmonary infections in cystic fibrosis

Clinical trials have included children and adolescents aged 5-17 years. More limited experience is available in treating children between 1 and 5 years of age.

Complicated urinary tract infections and pyelonephritis

Ciprofloxacin treatment of urinary tract infections should be considered when other treatments cannot be used, and should be based on the results of the microbiological documentation. Clinical trials have included children and adolescents aged 1-17 years.

Other specific severe infections

Other severe infections in accordance with official guidance, or after careful benefit-risk evaluation when other treatments cannot be used, or after failure to conventional therapy and when the

microbiological documentation can justify a ciprofloxacin use.

The use of ciprofloxacin for specific severe infections other than those mentioned above has not been evaluated in clinical trials and the clinical experience is limited. Consequently, caution is advised when treating patients with these infections.

Hypersensitivity

Hypersensitivity and allergic reactions, including anaphylaxis and anaphylactoid reactions, may occur following a single dose (see section 3.8) and may be life-threatening. If such reaction occurs, ciprofloxacin should be discontinued and an adequate medical treatment is required.

Tendinitis and tendon rupture

Tendinitis and tendon rupture (especially but not limited to Achilles tendon), sometimes bilateral, may occur as early as within 48 hours of starting treatment with fluoroquinolones and have been reported to occur even up to several months after discontinuation of treatment. The risk of tendinitis and tendon rupture is increased in older patients (above 60 years of age), with renal impairment, patients with solid organ transplants, and those treated concurrently with corticosteroids. Therefore, concomitant use of corticosteroids should be avoided.

At the first sign of tendinitis (e.g. painful swelling, inflammation) the treatment with ciprofloxacin should be discontinued and alternative treatment should be considered. The affected limb(s) should be appropriately treated (e.g. immobilisation). Corticosteroids should not be used if signs of tendinopathy occur.

Aortic aneurysm and dissection

Epidemiologic studies report an increased risk of aortic aneurysm and dissection after intake of fluoroquinolones, particularly in the older population. Therefore, fluoroquinolones should only be used after careful benefit-risk assessment and after consideration of other therapeutic options in patients with positive family history of aneurysm disease, or in patients diagnosed with pre-existing aortic aneurysm and/or aortic dissection, or in presence of other risk factors or conditions predisposing for aortic aneurysm and dissection (e.g. Marfan syndrome, vascular Ehlers-Danlos syndrome, Takayasu arteritis, giant cell arteritis, Behcet's disease, hypertension, known atherosclerosis). In case of sudden abdominal, chest or back pain, patients should be advised to immediately consult a physician in an emergency department.

Exacerbation of myasthenia gravis

Ciprofloxacin should be used with caution in patients with myasthenia gravis because symptoms can be exacerbated (see section 3.8).

Fluoroquinolones have neuromuscular blocking activity and may exacerbate muscle weakness in person with myasthenia gravis. Post marketing serious adverse events, including deaths and requirement for ventilator support have been associated with fluoroquinolones use in persons with myasthenia gravis. Avoid fluoroquinolones in patients with known history of myasthenia gravis.

Peripheral neuropathy

Cases of sensory or sensorimotor polyneuropathy resulting in paraesthesia, hypaesthesia, dysesthesia, or weakness have been reported in patients receiving quinolones and fluoroquinolones. Patients under treatment with ciprofloxacin should be advised to inform their doctor and pharmacist prior to continuing treatment if symptoms of neuropathy such as pain, burning, tingling, numbness, or weakness develop in order to prevent the development of potentially irreversible condition (see section Adverse Effects/Undesirable Effects).

Photosensitivity

Ciprofloxacin has been shown to cause photosensitivity reactions. Patients taking ciprofloxacin should

be advised to avoid direct exposure to either extensive sunlight or UV irradiation during treatment (see section 3.8).

Prolonged, disabling and potentially irreversible serious adverse drug reactions

Very rare cases of prolonged (continuing months or years), disabling and potentially irreversible serious adverse drug reactions affecting different, sometimes multiple body systems (musculoskeletal, nervous, psychiatric and senses) have been reported in patients receiving fluoroquinolones irrespective of their age and pre-existing risk factors. Ciprofloxacin should be discontinued immediately at the first signs or symptoms of any serious adverse reaction and patients should be advised to contact their prescriber for advice.

Central Nervous System

Ciprofloxacin like other quinolones are known to trigger seizures or lower the seizure threshold. Cases of status epilepticus have been reported. Ciprofloxacin should be used with caution in patients with CNS disorders which may be predisposed to seizure. If seizures occur ciprofloxacin should be discontinued (see section 3.8).

Psychiatric reactions may occur even after first administration of fluoroquinolones, including Ciprinol. In rare cases, depression or psychosis can progress to suicidal ideations/thoughts and self-injurious behaviour, such as attempted or completed suicide (see section 'Undesirable effects'). In the occurrence of such cases, Ciprinol should be discontinued and appropriate measures instituted. Caution is recommended if Ciprinol is to be used in psychotic patients or in patients with a history of psychiatric disease.

Cases of polyneuropathy (based on neurological symptoms such as pain, burning, sensory disturbances or muscle weakness, alone or in combination) have been reported in patients receiving ciprofloxacin. Ciprofloxacin should be discontinued in patients experiencing symptoms of neuropathy, including pain, burning, tingling, numbness, and/or weakness in order to prevent the development of an irreversible condition (see section 3.8).

Cardiac disorders

Caution should be taken when using fluoroquinolones, including Ciprofloxacin, in patients with known risk factors for prolongation of the QT interval such as, for example:

- congenital long QT syndrome
- concomitant use of drugs that are known to prolong the QT interval (e.g. Class IA and III anti-arrhythmics, tricyclic antidepressants, macrolides, antipsychotics)
- uncorrected electrolyte imbalance (e.g. hypokalaemia, hypomagnesaemia)
- cardiac disease (e.g. heart failure, myocardial infarction, bradycardia)

Elderly patients and women may be more sensitive to QTc-prolonging medications. Therefore, caution should be taken when using fluoroquinolones, including ciprofloxacin, in these populations. (See section 3.2 Geriatric patients, section 3.5, section 3.8, section 3.9).

Gastrointestinal System

The occurrence of severe and persistent diarrhoea during or after treatment (including several weeks after treatment) may indicate an antibiotic-associated colitis (life-threatening with possible fatal outcome), requiring immediate treatment (see section 3.8). In such cases, ciprofloxacin should immediately be discontinued, and an appropriate therapy initiated. Anti-peristaltic drugs are contraindicated in this situation.

Renal and urinary system

Crystalluria related to the use of ciprofloxacin has been reported (see section 3.8). Patients receiving ciprofloxacin should be well hydrated and excessive alkalinity of the urine should be avoided.

Impaired renal function

Since ciprofloxacin is largely excreted unchanged via renal pathway dose adjustment is needed in

patients with impaired renal function as described in section 4.2 to avoid an increase in adverse drug reactions due to accumulation of ciprofloxacin.

Hepatobiliary system

Cases of hepatic necrosis and life-threatening hepatic failure have been reported with ciprofloxacin (see section 3.8). In the event of any signs and symptoms of hepatic disease (such as anorexia, jaundice, dark urine, pruritus, or tender abdomen), treatment should be discontinued.

Glucose-6-phosphate dehydrogenase deficiency

Haemolytic reactions have been reported with ciprofloxacin in patients with glucose-6-phosphate dehydrogenase deficiency. Ciprofloxacin should be avoided in these patients unless the potential benefit is considered to outweigh the possible risk. In this case, potential occurrence of haemolysis should be monitored.

Resistance

During or following a course of treatment with ciprofloxacin bacteria that demonstrate resistance to ciprofloxacin may be isolated, with or without a clinically apparent superinfection. There may be a particular risk of selecting for ciprofloxacin-resistant bacteria during extended durations of treatment and when treating nosocomial infections and/or infections caused by *Staphylococcus* and *Pseudomonas* species.

Cytochrome P450

Ciprofloxacin inhibits CYP1A2 and thus may cause increased serum concentration of concomitantly administered substances metabolised by this enzyme (e.g. theophylline, clozapine, olanzapine, ropinirole, tizanidine, duloxetine). Co-administration of ciprofloxacin and tizanidine is contraindicated. Therefore, patients taking these substances concomitantly with ciprofloxacin should be monitored closely for clinical signs of overdose, and determination of serum concentrations (e.g. of theophylline) may be necessary (see section 3.5).

Dysglycemia

As with all fluoroquinolones, disturbances in blood glucose, including both hypoglycemia and hyperglycemia have been reported with Ciprinol. In Ciprinol-treated patients, dysglycemia occurred predominantly in elderly diabetic patients receiving concomitant treatment with an oral hypoglycemic agent (e.g. sulfonylurea) or with insulin. Cases of hypoglycaemic coma have been reported. In diabetic patients, careful monitoring of blood glucose is recommended (see section 3.8).

Methotrexate

The concomitant use of ciprofloxacin with methotrexate is not recommended (see section 3.5).

Interaction with tests

The *in-vitro* activity of ciprofloxacin against *Mycobacterium tuberculosis* might give false negative bacteriological test results in specimens from patients currently taking ciprofloxacin.

Effects on the ability to drive and use machines

Due to its neurological effects, ciprofloxacin may affect reaction time. Thus, the ability to drive or to operate machinery may be impaired.

3.5 Interaction with other medicinal products and other forms of interaction

Effects of other products on ciprofloxacin:

Drugs known to prolong QT interval

Ciprofloxacin, like other fluoroquinolones, should be used with caution in patients receiving drugs

known to prolong QT interval (e.g. Class IA and III anti-arrhythmics, tricyclic antidepressants, macrolides, antipsychotics) (see section 3.4).

Chelation Complex Formation

The simultaneous administration of ciprofloxacin (oral) and multivalent cation-containing drugs and mineral supplements (e.g. calcium, magnesium, aluminium, iron), polymeric phosphate binders (e.g. sevelamer), sucralfate or antacids, and highly buffered drugs (e.g. didanosine tablets) containing magnesium, aluminium, or calcium reduces the absorption of ciprofloxacin. Consequently, ciprofloxacin should be administered either 1-2 hours before or at least 4 hours after these preparations.

The restriction does not apply to antacids belonging to the class of H₂ receptor blockers.

Food and Dairy Products

Dietary calcium as part of a meal does not significantly affect absorption. However, the concurrent administration of dairy products or mineral-fortified drinks alone (e.g. milk, yoghurt, calcium-fortified orange juice) with ciprofloxacin should be avoided because absorption of ciprofloxacin may be reduced.

Probenecid

Probenecid interferes with renal secretion of ciprofloxacin. Co-administration of probenecid and ciprofloxacin increases ciprofloxacin serum concentrations.

Metoclopramide

Metoclopramide accelerates the absorption of ciprofloxacin (oral) resulting in a shorter time to reach maximum plasma concentrations. No effect was seen on the bioavailability of ciprofloxacin.

Omeprazole

Concomitant administration of ciprofloxacin and omeprazole containing medicinal products results in a slight reduction of C_{max} and AUC of ciprofloxacin.

Effects of ciprofloxacin on other medicinal products:

Tizanidine

Tizanidine must not be administered together with ciprofloxacin. There was an increase in serum tizanidine concentration when given concomitantly with ciprofloxacin. Increased serum tizanidine concentration is associated with a potentiated hypotensive and sedative effect.

Methotrexate

Renal tubular transport of methotrexate may be inhibited by concomitant administration of ciprofloxacin, potentially leading to increased plasma levels of methotrexate and increased risk of methotrexate-associated toxic reactions. The concomitant use is not recommended (see section 3.4).

Theophylline

Concurrent administration of ciprofloxacin and theophylline can cause an undesirable increase in serum theophylline concentration. This can lead to theophylline-induced side effects that may rarely be life threatening or fatal. During the combination, serum theophylline concentrations should be checked and the theophylline dose reduced as necessary (see section 3.4).

Other xanthine derivatives

On concurrent administration of ciprofloxacin and caffeine or pentoxifylline (oxpentifylline), raised serum concentrations of these xanthine derivatives were reported.

Phenytoin

Simultaneous administration of ciprofloxacin and phenytoin may result in increased or reduced serum

levels of phenytoin such that monitoring of drug levels is recommended.

Cyclosporin

A transient rise in the concentration of serum creatinine was observed when ciprofloxacin and cyclosporin containing medicinal products were administered simultaneously. Therefore, it is frequently (twice a week) necessary to control the serum creatinine concentrations in these patients.

Vitamin K antagonists

Simultaneous administration of ciprofloxacin with a vitamin K antagonist may augment its anticoagulant effects. The risk may vary with the underlying infection, age and general status of the patient so that the contribution of ciprofloxacin to the increase in INR (international normalized ratio) is difficult to assess. The INR should be monitored frequently during and shortly after co-administration of ciprofloxacin with a vitamin K antagonist (e.g., warfarin, acenocoumarol, phenprocoumon, or fluindione).

Duloxetine

Concomitant use of duloxetine with strong inhibitors of the CYP450 1A2 isozyme such as fluvoxamine, may result in an increase of AUC and C_{max} of duloxetine. Although no clinical data are available on a possible interaction with ciprofloxacin, similar effects can be expected upon concomitant administration (see section 3.4).

Ropinirole

Concomitant use of ropinirole with ciprofloxacin, a moderate inhibitor of the CYP450 1A2 isozyme, results in an increase of C_{max} and AUC of ropinirole. Monitoring of ropinirole-related side effects and dose adjustment as appropriate is recommended during and shortly after co-administration with ciprofloxacin (see section 3.4).

Lidocaine

Concomitant use of lidocaine containing medicinal products with ciprofloxacin, a moderate inhibitor of CYP450 1A2 isozyme, reduces clearance of intravenous lidocaine. Although lidocaine treatment was well tolerated, a possible interaction with ciprofloxacin associated with side effects may occur upon concomitant administration.

Clozapine

Following concomitant administration of 250 mg ciprofloxacin with clozapine for 7 days, serum concentrations of clozapine and N-desmethylozapine were increased. Clinical surveillance and appropriate adjustment of clozapine dosage during and shortly after co-administration with ciprofloxacin are advised (see section 3.4).

Sildenafil

C_{max} and AUC of sildenafil were increased approximately twofold after an oral dose of 50 mg given concomitantly with 500 mg ciprofloxacin. Therefore, caution should be used prescribing ciprofloxacin concomitantly with sildenafil taking into consideration the risks and the benefits.

Agomelatine

Fluvoxamine, as a strong inhibitor of the CYP450 1A2 isoenzyme, markedly inhibits the metabolism of agomelatine resulting in a 60-fold increase of agomelatine exposure. Although no clinical data are available for a possible interaction with ciprofloxacin, a moderate inhibitor of CYP450 1A2, similar effects can be expected upon concomitant administration (see 'Cytochrome P450' in section 3.4).

Zolpidem

Co-administration of ciprofloxacin may increase blood levels of zolpidem, concurrent use is not

recommended.

3.6 Pregnancy and lactation

Pregnancy

The data that are available on administration of ciprofloxacin to pregnant women indicates no malformative or feto/neonatal toxicity of ciprofloxacin. Animal studies do not indicate direct or indirect harmful effects with respect to reproductive toxicity. In juvenile and prenatal animals exposed to quinolones, effects on immature cartilage have been observed, thus, it cannot be excluded that the drug could cause damage to articular cartilage in the human immature organism / foetus. As a precautionary measure, it is preferable to avoid the use of ciprofloxacin during pregnancy.

Lactation

Ciprofloxacin is excreted in breast milk. Due to the potential risk of articular damage, ciprofloxacin should not be used during breast-feeding.

3.7 Effects on ability to drive and use machines

Due to its neurological effects, ciprofloxacin may affect reaction time. Thus, the ability to drive or to operate machinery may be impaired.

3.8 Undesirable effects

The most commonly reported adverse drug reactions (ADRs) are nausea and diarrhoea.

System Organ Class	Common	Uncommon	Rare	Very Rare	Frequency not known
Infections and Infestations		Mycotic superinfections	Antibiotic associated colitis (very rarely with possible fatal outcome) (see section 4.4)		
Blood and Lymphatic System Disorders		Eosinophilia	Leukopenia Anaemia Neutropenia Leukocytosis Thrombocytopenia Thrombocythaemia	Haemolytic anaemia Agranulocytosis Pancytopenia (lifethreatening) Bone marrow depression (lifethreatening)	
Immune System Disorders			Allergic reaction Allergic oedema / angiooedema	Anaphylactic reaction Anaphylactic shock (lifethreatening) (see section 3.4) Serum sicknesslike reaction	
Endocrine disorders					Syndrome of inappropriate secretion of

System Organ Class	Common	Uncommon	Rare	Very Rare	Frequency not known
					antidiuretic hormone (SIADH)
Metabolism and Nutrition Disorders		Anorexia	Hyperglycaemia Hypoglycemia		Hypoglycaemic coma (see section 3.4)
Psychiatric Disorders*		Psychomotor hyperactivity / agitation	Confusion and disorientation Anxiety reaction Abnormal dreams Depression (potentially culminating in suicidal ideations/ thoughts or suicide attempts and completed suicide) (see section 3.4) Hallucinations	Psychotic reactions (potentially culminating in suicidal ideations/ thoughts or suicide attempts and completed suicide) (see section 3.4)	
Nervous System Disorders*		Headache Dizziness Sleep disorders Taste disorders	Par- and Dysaesthesia Hypoaesthesia Tremor Seizures (incl. status epilepticus see section 3.4) Vertigo	Migraine Disturbed coordination Gait disturbance Olfactory nerve disorders Intracranial hypertension	Peripheral neuropathy (see section 3.4)
Eye Disorders*			Visual disturbances (e.g. diplopia)	Visual colour distortions	
Ear and Labyrinth Disorders*			Tinnitus Hearing loss /	Hearing impaired	
Cardiac Disorders			Tachycardia		ventricular arrhythmia, torsades de pointes (reported predominantly in patients with risk factors for QT prolongation), ECG QT prolonged (see section 3.4 and 3.9)* * preliminary

System Organ Class	Common	Uncommon	Rare	Very Rare	Frequency not known
					wording according to the assessment report; will be aligned with final wording of PhVWP.
Vascular Disorders			Vasodilatation Hypotension Syncope	Vasculitis	
Respiratory, Thoracic and Mediastinal Disorders			Dyspnoea (including asthmatic condition)		
Gastrointestinal Disorders	Nausea Diarrhoea	Vomiting Gastrointestinal and abdominal pains Dyspepsia Flatulence		Pancreatitis	
Hepatobiliary Disorders		Increase in transaminases Increased bilirubin	Hepatic impairment Cholestatic icterus Hepatitis	Liver necrosis (very rarely progressing to life-threatening hepatic failure) (see section 3.4)	
Skin and Subcutaneous Tissue Disorders		Rash Pruritus Urticaria	Photosensitivity reactions (see section 3.4)	Petechiae Erythema multiforme Erythema nodosum Stevens-Johnson syndrome (potentially lifethreatening) Toxic epidermal necrolysis (potentially lifethreatening)	Acute generalised exanthematous pustulosis (AGEP)
Musculoskeletal, Connective Tissue and Bone Disorders*		Musculoskeletal pain (e.g. extremity pain, back pain, chest pain) Arthralgia	Myalgia Arthritis Increased muscle tone and cramping	Muscular weakness Tendinitis Tendon rupture (predominantly Achilles tendon) (see section 3.4) Exacerbation of symptoms of	

System Organ Class	Common	Uncommon	Rare	Very Rare	Frequency not known
				myasthenia gravis (see section 3.4)	
Renal and Urinary Disorders		Renal impairment	Renal failure Haematuria Crystalluria (see section 3.4) Tubulointerstitial nephritis		
General Disorders and Administration Site Conditions*		Asthenia Fever	Oedema Sweating (hyperhidrosis)		
Investigations		Increase in blood alkaline phosphatase	Increased amylase		International normalised ratio increased (in patients treated with Vitamin K antagonists)

*Very rare cases of prolonged (up to months or years), disabling and potentially irreversible serious drug reactions affecting several, sometimes multiple, system organ classes and senses (including reactions such as tendinitis, tendon rupture, arthralgia, pain in extremities, gait disturbance, neuropathies associated with paraesthesia, depression, fatigue, memory impairment, sleep disorders, and impairment of hearing, vision, taste and smell) have been reported in association with the use of fluoroquinolones in some cases irrespective of pre-existing risk factors (see section Warnings and Precautions).

Exacerbation of myasthenia gravis
Post Marketing Experience

Paediatric patients

In children, arthropathy is reported to occur commonly (see section 3.4).

3.9 Overdose

Based on the limited information available in two cases of ingestion of over 18g of ciprofloxacin, reversible renal toxicity has occurred.

Symptoms in overdose consist of dizziness, tremor, headache, tiredness, seizures, hallucinations, confusion, abdominal discomfort, renal and hepatic impairment as well as crystalluria and haematuria. Therefore, apart from routine emergency measures e.g. gastric emptying followed by medical carbon, it is recommended to monitor renal function, including urinary pH and acidify, if required, to prevent crystalluria. Patients must be kept well hydrated, and in the case of renal damage resulting in prolonged oliguria, dialysis should be initiated.

Calcium or magnesium antacids may be administered as soon as possible after ingestion of ciprofloxacin tablets in order to reduce the absorption of ciprofloxacin. Only a small quantity of ciprofloxacin (<10%) is eliminated by haemodialysis or peritoneal dialysis.

In the event of overdose, symptomatic treatment should be implemented. ECG monitoring should be

undertaken, because of the possibility of QT interval prolongation.

4. PHARMACOLOGICAL PROPERTIES

4.1 Pharmacodynamic properties

According to the Anatomical Therapeutic Chemical Classification (ATC), Ciprinol is classified into a group of fluoroquinolones (J01MA02).

Mechanism of action:

As a fluoroquinolone antibacterial agent, the bactericidal action of ciprofloxacin results from the inhibition of both type II topoisomerase (DNA-gyrase) and topoisomerase IV, required for bacterial DNA replication, transcription, repair and recombination.

PK/PD relationship:

Efficacy mainly depends on the relation between the maximum concentration in serum (C_{max}) and the minimum inhibitory concentration (MIC) of ciprofloxacin for a bacterial pathogen and the relation between the area under the curve (AUC) and the MIC.

Mechanism of resistance:

In-vitro resistance to ciprofloxacin can be acquired through a stepwise process by target site mutations in both DNA gyrase and topoisomerase IV. The degree of cross-resistance between ciprofloxacin and other fluoroquinolones that results is variable. Single mutations may not result in clinical resistance, but multiple mutations generally result in clinical resistance to many or all active substances within the class.

Impermeability and/or active substance efflux pump mechanisms of resistance may have a variable effect on susceptibility to fluoroquinolones, which depends on the physiochemical properties of the various active substances within the class and the affinity of transport systems for each active substance. All in-vitro mechanisms of resistance are commonly observed in clinical isolates. Resistance mechanisms that inactivate other antibiotics such as permeation barriers (common in *Pseudomonas aeruginosa*) and efflux mechanisms may affect susceptibility to ciprofloxacin. Plasmid-mediated resistance encoded by qnr-genes has been reported.

Spectrum of antibacterial activity:

Breakpoints separate susceptible strains from strains with intermediate susceptibility and the latter from resistant strains:

EUCAST Recommendations

Microorganisms	Susceptible	Resistant
<i>Enterobacteria</i>	S ≤ 0.5 mg/l	R > 1 mg/l
<i>Pseudomonas</i>	S ≤ 0.5 mg/l	R > 1 mg/l
<i>Acinetobacter</i>	S ≤ 1 mg/l	R > 1 mg/l
<i>Staphylococcus</i> spp.	S ≤ 1 mg/l	R > 1 mg/l
<i>Haemophilus influenzae</i> and <i>Moraxella catarrhalis</i>	S ≤ 0.5 mg/l	R > 0,5 mg/l
<i>Neisseria gonorrhoeae</i>	S ≤ 0.03 mg/l	R > 0,06 mg/l
<i>Neisseria meningitidis</i>	S ≤ 0.03 mg/l	R > 0,06 mg/l
Non-species-related breakpoints*	S ≤ 0.5 mg/l	R > 1 mg/l

1 *Staphylococcus* spp. - breakpoints for ciprofloxacin relate to high dose therapy.

* Non-species-related breakpoints have been determined mainly on the basis of PK/PD data and are

independent of MIC distributions of specific species. They are for use only for species that have not been given a species-specific breakpoint and not for those species where susceptibility testing is not recommended.

The prevalence of acquired resistance may vary geographically and with time for selected species and local information on resistance is desirable, particularly when treating severe infections. As necessary, expert advice should be sought when the local prevalence of resistance is such that the utility of the agent in at least some types of infections is questionable.

Groupings of relevant species according to ciprofloxacin susceptibility (for *Streptococcus* species see section 3.4)

COMMONLY SUSCEPTIBLE SPECIES
Aerobic Gram-positive micro-organisms <i>Bacillus anthracis</i> (1)
Aerobic Gram-negative micro-organisms <i>Aeromonas</i> spp. <i>Brucella</i> spp. <i>Citrobacter koseri</i> <i>Francisella tularensis</i> <i>Haemophilus ducreyi</i> <i>Haemophilus influenzae</i> * <i>Legionella</i> spp. <i>Moraxella catarrhalis</i> * <i>Neisseria meningitidis</i> <i>Pasteurella</i> spp. <i>Salmonella</i> spp.* <i>Shigella</i> spp.* <i>Vibrio</i> spp. <i>Yersinia pestis</i>
Anaerobic micro-organisms <i>Mobiluncus</i>
Other micro-organisms <i>Chlamydia trachomatis</i> (\$) <i>Chlamydia pneumoniae</i> (\$) <i>Mycoplasma hominis</i> (\$) <i>Mycoplasma pneumoniae</i> (\$)
SPECIES FOR WHICH ACQUIRED RESISTANCE MAY BE A PROBLEM
Aerobic Gram-positive micro-organisms <i>Enterococcus faecalis</i> (\$) <i>Staphylococcus</i> spp.* (2)
Aerobic Gram-negative micro-organisms <i>Acinetobacter baumannii</i> + <i>Burkholderia cepacia</i> +* <i>Campylobacter</i> spp.+* <i>Citrobacter freundii</i> * <i>Enterobacter aerogenes</i> <i>Enterobacter cloacae</i> * <i>Escherichia coli</i> * <i>Klebsiella oxytoca</i> <i>Klebsiella pneumoniae</i> * <i>Morganella morganii</i> * <i>Neisseria gonorrhoeae</i> *

<i>Proteus mirabilis</i> * <i>Proteus vulgaris</i> * <i>Providencia</i> spp. <i>Pseudomonas aeruginosa</i> * <i>Pseudomonas fluorescens</i> <i>Serratia marcescens</i> *
Anaerobic micro-organisms <i>Peptostreptococcus</i> spp. <i>Propionibacterium acnes</i>
INHERENTLY RESISTANT ORGANISMS
Aerobic Gram-positive micro-organisms <i>Actinomyces</i> <i>Enterococcus faecium</i> <i>Listeria monocytogenes</i>
Aerobic Gram-negative micro-organisms <i>Stenotrophomonas maltophilia</i>
Anaerobic micro-organisms <i>Excepted as listed above</i>
Other micro-organisms <i>Mycoplasma genitalium</i> <i>Ureaplasma urealyticum</i>
* Clinical efficacy has been demonstrated for susceptible isolates in approved clinical indications + Resistance rate $\geq 50\%$ in one or more EU countries (\$): Natural intermediate susceptibility in the absence of acquired mechanism of resistance (1): Studies have been conducted in experimental animal infections due to inhalations of <i>Bacillus anthracis</i> spores; these studies reveal that antibiotics starting early after exposition avoid the occurrence of the disease if the treatment is made up to the decrease of the number of spores in the organism under the infective dose. The recommended use in human subjects is based primarily on <i>in-vitro</i> susceptibility and on animal experimental data together with limited human data. Two-month treatment duration in adults with oral ciprofloxacin given at the following dose, 500 mg bid, is considered as effective to prevent anthrax infection in humans. The treating physician should refer to national and/or international consensus documents regarding treatment of anthrax. (2): Methicillin-resistant <i>S. aureus</i> very commonly express co-resistance to fluoroquinolones. The rate of resistance to methicillin is around 20 to 50% among all staphylococcal species and is usually higher in nosocomial isolates.

4.2 Pharmacokinetic properties

Absorption

Following oral administration of single doses of 250 mg, 500 mg, and 750 mg of ciprofloxacin tablets, ciprofloxacin is absorbed rapidly and extensively, mainly from the small intestine, reaching maximum serum concentrations 1-2 hours later.

Single doses of 100-750 mg produced dose-dependent maximum serum concentrations (C_{max}) between 0.56 and 3.7 mg/l. Serum concentrations increase proportionately with doses up to 1000 mg. The absolute bioavailability is approximately 70-80%.

A 500 mg oral dose given every 12 hours has been shown to produce an area under the serum concentration-time curve (AUC) equivalent to that produced by an intravenous infusion of 400 mg ciprofloxacin given over 60 minutes every 12 hours.

Distribution

Protein binding of ciprofloxacin is low (20-30%). Ciprofloxacin is present in plasma largely in a nonionised form and has a large steady state distribution volume of 2-3 l/kg body weight.

Ciprofloxacin reaches high concentrations in a variety of tissues such as lung (epithelial fluid, alveolar

macrophages, biopsy tissue), sinuses, inflamed lesions (cantharides blister fluid), and the urogenital tract (urine, prostate, endometrium) where total concentrations exceeding those of plasma concentrations are reached.

Metabolism

Low concentrations of four metabolites have been reported, which were identified as: desethyleneciprofloxacin (M₁), sulphociprofloxacin (M₂), oxociprofloxacin (M₃) and formylciprofloxacin (M₄). The metabolites display *in-vitro* antimicrobial activity but to a lower degree than the parent compound.

Ciprofloxacin is known to be a moderate inhibitor of the CYP 450 1A2 iso-enzymes.

Elimination

Ciprofloxacin is largely excreted unchanged both renally and, to a smaller extent, faecally. The serum elimination half-life in subjects with normal renal function is approximately 4-7 hours.

Excretion of ciprofloxacin (% of dose)		
	Oral administration	
	Urine	Feces
Ciprofloxacin	44.7	25.0
Metabolites (M ₁ -M ₄)	11.3	7.5

Renal clearance is between 180-300 mL/kg/h and the total body clearance is between 480-600 mL/kg/h. Ciprofloxacin undergoes both glomerular filtration and tubular secretion. Severely impaired renal function leads to increased half lives of ciprofloxacin of up to 12 h.

Non-renal clearance of ciprofloxacin is mainly due to active trans-intestinal secretion and metabolism. 1 % of the dose is excreted via the biliary route. Ciprofloxacin is present in the bile in high concentrations.

Pediatric patients

The pharmacokinetic data in paediatric patients are limited.

In a study in children C_{max} and AUC were not age-dependent (above one year of age). No notable increase in C_{max} and AUC upon multiple dosing (10 mg/kg three times daily) was observed.

In 10 children with severe sepsis C_{max} was 6.1 mg/l (range 4.6-8.3 mg/l) after a 1-hour intravenous infusion of 10 mg/kg in children aged less than 1 year compared to 7.2 mg/l (range 4.7-11.8 mg/l) for children between 1 and 5 years of age. The AUC values were 17.4 mg*h/L (range 11.8-32.0 mg*h/l) and 16.5 mg*h/l (range 11.0-23.8 mg*h/l) in the respective age groups.

These values are within the range reported for adults at therapeutic doses. Based on population pharmacokinetic analysis of pediatric patients with various infections, the predicted mean half-life in children is approx. 4-5 hours and the bioavailability of the oral suspension ranges from 50 to 80%.

5. PHARMACEUTICAL PARTICULARS

5.1 Special precautions for storage

Do not store above 30 °C.

Store in the original package in order to protect from light and moisture.

5.2 Nature and contents of container

Ciprinol is available in boxes of 10 and 100 film-coated tablets in blister packs.

6. PRODUCT REGISTRATION HOLDER

PAHANG PHARMACY SDN. BHD.,
Lot 5979, Jalan Teratai, 5 ½ Miles, Off Jalan Meru,
41050 Klang, Selangor, Malaysia

7. NAME AND ADDRESS OF THE MANUFACTURER

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